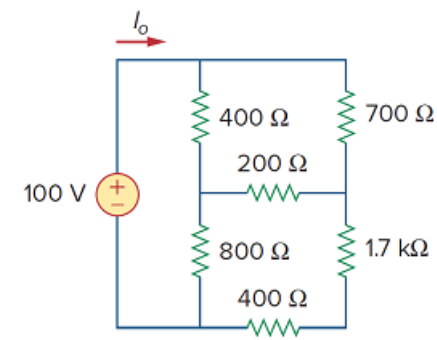


Problem

Calculate I_o in the circuit of Fig. 2.119.

Figure 2.119:



Step-by-step solution

Step 1 of 4

Consider the textbook Figure 2.119. The upper resistors $400\ \Omega$, $700\ \Omega$ and $200\ \Omega$ form a Δ circuit. Transform the Δ circuit to Y circuit and the resultant network is as shown in Figure 1.

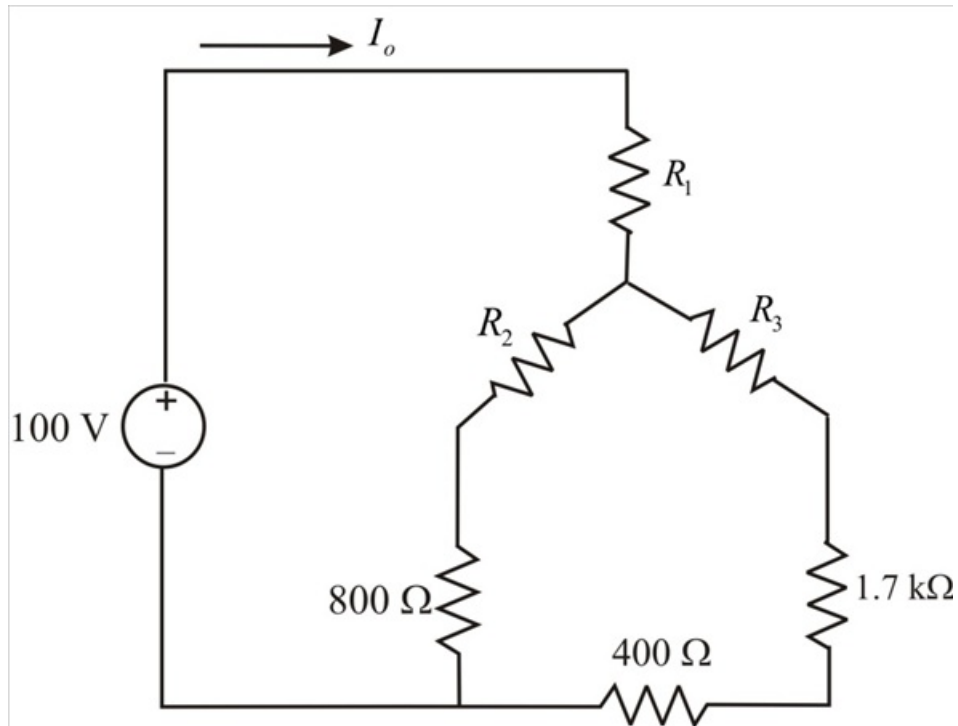


Figure 1

Step 2 of 4

Calculate the resistance R_1 .

$$\begin{aligned} R_1 &= \frac{(400)(700)}{400 + 700 + 200} \\ &= \frac{(400)(700)}{1300} \\ &= 215.4 \, \Omega \end{aligned}$$

Calculate the resistance R_2 .

$$\begin{aligned} R_2 &= \frac{(400)(200)}{400 + 700 + 200} \\ &= \frac{(400)(200)}{1300} \\ &= 61.5 \, \Omega \end{aligned}$$

Step 3 of 4

Calculate the resistance R_3 .

$$\begin{aligned} R_3 &= \frac{(700)(200)}{400 + 700 + 200} \\ &= \frac{(700)(200)}{1300} \\ &= 107.7 \, \Omega \end{aligned}$$

Calculate the equivalent resistance.

$$\begin{aligned} R_{\text{eq}} &= R_1 + [(R_2 + 800) \parallel (R_3 + 1.7\text{k} + 400)] \\ &= 215.4 + [(861.5) \parallel (2207.7)] \\ &= 215.4 + \left[\frac{(861.5)(2207.7)}{3069.2} \right] \\ &= 835 \, \Omega \end{aligned}$$

Step 4 of 4

Calculate the current I_o .

$$\begin{aligned} I_o &= \frac{V}{R_{\text{eq}}} \\ &= \frac{100}{835} \\ &= 0.12 \, \text{A} \end{aligned}$$

Therefore, the current I_o of the textbook Figure 2.119 is 0.12 A.